AI COURSE - II AIML/REC
CATEGORY 8: KNOWLEDGE BASE SYSTEM

Topics:
1. Medical Diagnosis
2. Autonomous Vehicle
3. Industrial Manufacturing

Using Facts, Rules, Forward Chaining and Backward Chaining.

Common Forward Chaining Function
def forward_chaining(facts, rules): ...

Common Backward Chaining Function
def backward_chaining(goal, facts, rules): ...

The document includes the complete Python program for Medical Diagnosis, Autonomous Vehicle, and Industrial Manufacturing expert-system examples using forward and backward chaining as provided by the user.

```
MEDICAL DIAGNOSIS: Facts = {fever, cough, headache, body_pain}; Rules derive flu,
common_cold, malaria, viral_infection.
AUTONOMOUS VEHICLE: Facts = {obstacle_detected, high_speed, traffic_signal_red};
Rules derive apply_brake, reduce_speed, stop_vehicle, collision_risk.
INDUSTRIAL MANUFACTURING: Facts = {raw_material_available, machine_ready,
power_available, quality_check_passed}; Rules derive production_started,
product_manufactured, product_approved, ready_for_delivery.

Main program executes all three systems.
```